

Q.No:9

DS	Session	Searching	Question Information	Level 1	Challenge 9
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Problem Description:

Prabhu Salamon is planning to make a very long journey across the cityside by Train. His journey consists of N train routes, numbered from 1 to N in the order he must take them. The trains themselves are very fast, but do not run often. The i th train route only runs every X_i days.

More specifically, he can only take the i th train on day X_i , $2X_i$, $3X_i$ and so on. Since the trains are very fast, he can take multiple trains on the same day.

Prabhu Salamon must finish his journey by day D , but he would like to start the journey as late as possible. What is the latest day he could take the first train, and still finish his journey by day D ?

It is guaranteed that it is possible for Prabhu Salamon to finish his journey by day D .

Constraints:

$1 \leq T \leq 100$.

$1 \leq X_i \leq D$.

$1 \leq N \leq 1000$.

$1 \leq D \leq 10^{12}$

Input Format:

The first line of the input gives the number of test cases, T . T test cases follow. Each test case begins with a line containing the two integers N and D . Then, another line follows containing N integers, the i th one is X_i .

Output Format:

Print the output in a single line contains, the latest day he could take the first train, and still finish his journey by day D .

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
3
3 100
```

Test Case 2

INPUT (STDIN)

```
3
5 1000
```

```
31 71 21
4 100
11 16 51 50
1 1
1
```

EXPECTED OUTPUT

```
62
44
1
```

```
311 171 211 45 32
6 1000
111 161 511 510 56 93
2 10
9 5
```

EXPECTED OUTPUT

```
622
0
9
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
for(int t=0;t<T;t++)
```

Test Case 2

KEYWORD

```
for(int i=n-1;i>=0;i--)
```

Complexity Test Cases

Code:

```
#include <stdio.h>

int main()
{
    int T;
    scanf("%d", &T);

    for (int t = 0; t < T; t++)
```

```
{
int N;
long long D;

scanf("%d %lld", &N, &D);

long long x[1000];

for (int i = 0; i < N; i++)
{
scanf("%lld", &x[i]);
}
for (int i = N - 1; i >= 0; i--)
{
D = (D / x[i]) * x[i];
}

printf("%lld\n", D);
}

return 0;
}
```

Output:

Test_case 1;

```
● aaditya.jaiswal@Aadityas-MacBook-Air: Search...
3
3 100
31 71 21
62
4 100
11 16 51 50
44
1 1
1
1
```

Test_case 2:

```
3
5 1000
311 171 211 45 32
622
6 1000
111 161 511 510 56 93
0
2 10
9 5
9
```